

PHW251 Problem Set 2

Answer Key

2026-09-29

Render this file from the `problem_set_2` folder so `co2_mm_mlo.csv` is found.

Question 1

Use `Sys.Date()` and save the result.

```
today <- Sys.Date()  
today
```

```
[1] "2026-09-29"
```

Check the class and the storage type.

```
class(today)
```

```
[1] "Date"
```

```
typeof(today)
```

```
[1] "double"
```

```
as.numeric(today)
```

```
[1] 20725
```

Why is the object both a Date and a double?

Dates in R are stored as the number of days before or after 1 January 1970. That count is an integer stored as a double, which is why `typeof()` returns "double" and `as.numeric()` prints a large number of days. Because the value came from `Sys.Date()`, R also keeps the Date class and prints it as a calendar date instead of as that number.

`wday()` returns an abbreviated weekday when `label = TRUE`. The default for `abbr` is already `TRUE`, so the result looks like "Sun" or "Mon".

```
library(lubridate)
```

```
Attaching package: 'lubridate'
```

The following objects are masked from 'package:base':

```
date, intersect, setdiff, union
```

```
current_wday <- wday(today, label = TRUE)
current_wday
```

```
[1] Tue
Levels: Sun < Mon < Tue < Wed < Thu < Fri < Sat
```

```
days <- levels(wday(Sys.Date(), label = TRUE))
days
```

```
[1] "Sun" "Mon" "Tue" "Wed" "Thu" "Fri" "Sat"
```

Those levels start on Sunday, so Monday through Friday are positions 2 through 6.

```
weekdays <- days[2:6]
weekdays
```

```
[1] "Mon" "Tue" "Wed" "Thu" "Fri"
```

```
current_wday %in% weekdays
```

```
[1] TRUE
```

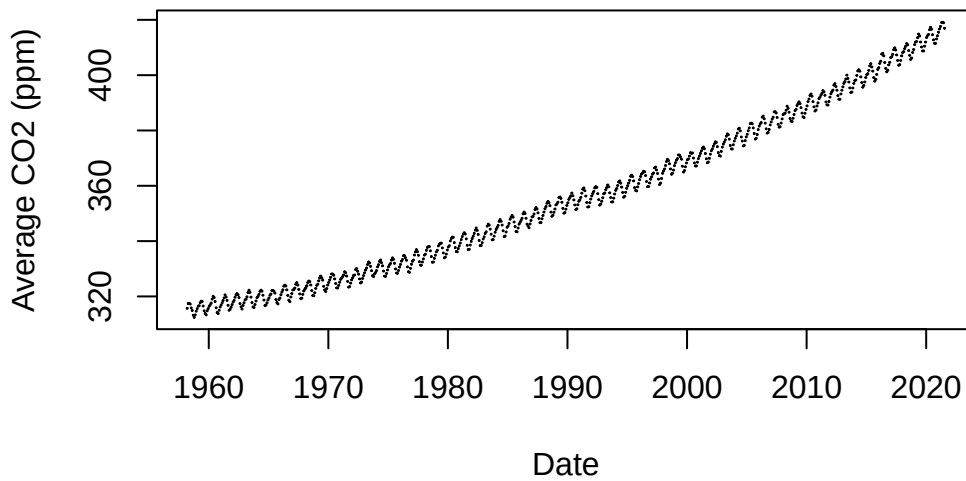
```
current_wday %in% days
```

```
[1] TRUE
```

The second check is always `TRUE`, because `current_wday` is one of the seven day names. The first check is `TRUE` on Monday through Friday and `FALSE` on Saturday or Sunday.

Question 2

```
co2 <- read.csv(file = "co2_mm_mlo.csv", skip = 51, header = TRUE, row.names = NULL)
colnames(co2) <- c("year", "month", "date", "average", "interpolated", "trend", "ndays")
plot(x = co2$date, y = co2$average, pch = 20, cex = 0.1,
     xlab = "Date", ylab = "Average CO2 (ppm)")
```



```
tail(co2)
```

	year	month	date	average	interpolated	trend	ndays	NA
756	2021	2	2021.125	416.75	415.93	28	1.02	0.37
757	2021	3	2021.208	417.64	416.14	28	0.86	0.31
758	2021	4	2021.292	419.05	416.23	24	1.12	0.44
759	2021	5	2021.375	419.13	415.71	28	0.90	0.33
760	2021	6	2021.458	418.94	416.58	28	0.65	0.23
761	2021	7	2021.542	416.96	416.58	30	0.72	0.25

```
avg_co2 <- co2$average
avg_co2 <- co2[["average"]]
```

```

[1] 315.70 317.45 317.51 317.24 315.86 314.93 313.20 312.43 313.33 314.67
[11] 315.58 316.48 316.65 317.72 318.29 318.15 316.54 314.80 313.84 313.33
[21] 314.81 315.58 316.43 316.98 317.58 319.03 320.04 319.59 318.18 315.90
[31] 314.17 313.83 315.00 316.19 316.89 317.70 318.54 319.48 320.58 319.77
[41] 318.57 316.79 314.99 315.31 316.10 317.01 317.94 318.55 319.68 320.57
[51] 321.02 320.62 319.61 317.40 316.25 315.42 316.69 317.70 318.74 319.07
[61] 319.86 321.38 322.25 321.48 319.74 317.77 316.21 315.99 317.07 318.35
[71] 319.57 320.01 320.74 321.84 322.26 321.89 320.44 318.69 316.70 316.87
[81] 317.68 318.71 319.44 320.44 320.89 322.14 322.17 321.87 321.21 318.87
[91] 317.81 317.30 318.87 319.42 320.62 321.60 322.39 323.70 324.08 323.75
[101] 322.38 320.36 318.64 318.10 319.78 321.03 322.33 322.50 323.04 324.42
[111] 325.00 324.09 322.54 320.92 319.25 319.39 320.73 321.96 322.57 323.15
[121] 323.89 325.02 325.57 325.36 324.14 322.11 320.33 320.25 321.32 322.89
[131] 324.00 324.42 325.63 326.66 327.38 326.71 325.88 323.66 322.38 321.78
[141] 322.86 324.12 325.06 325.98 326.93 328.13 328.08 327.67 326.34 324.69
[151] 323.10 323.06 324.01 325.13 326.17 326.68 327.17 327.79 328.93 328.57
[161] 327.36 325.43 323.36 323.56 324.80 326.01 326.77 327.63 327.75 329.72
[171] 330.07 329.09 328.04 326.32 324.84 325.20 326.50 327.55 328.55 329.56
[181] 330.30 331.50 332.48 332.07 330.87 329.31 327.51 327.18 328.16 328.64
[191] 329.35 330.71 331.48 332.65 333.19 332.20 331.07 329.15 327.33 327.28
[201] 328.31 329.58 330.73 331.46 331.94 333.11 333.95 333.42 331.97 329.95
[211] 328.50 328.36 329.38 330.62 331.56 332.74 333.36 334.74 334.72 333.98
[221] 333.08 330.68 328.96 328.72 330.16 331.62 332.68 333.17 334.96 336.14
[231] 336.93 336.17 334.88 332.56 331.29 331.28 332.46 333.60 334.94 335.26
[241] 336.66 337.69 338.02 338.01 336.50 334.42 332.36 332.45 333.76 334.91
[251] 336.14 336.69 338.27 338.82 339.24 339.26 337.54 335.72 333.98 334.24
[261] 335.32 336.81 337.90 338.34 340.07 340.93 341.45 341.36 339.45 337.67
[271] 336.25 336.14 337.30 338.29 339.29 340.55 341.63 342.60 343.04 342.54
[281] 340.82 338.48 336.95 337.05 338.58 339.91 340.93 341.76 342.78 343.96
[291] 344.77 343.88 342.42 340.24 338.38 338.41 339.44 340.78 341.57 342.78
[301] 343.37 345.40 346.14 345.76 344.32 342.50 340.46 340.53 341.79 343.20
[311] 344.21 344.92 345.68 347.14 347.78 347.16 345.79 343.74 341.59 341.86
[321] 343.31 345.00 345.48 346.41 347.91 348.66 349.28 348.65 346.90 345.26
[331] 343.47 343.35 344.73 346.12 346.78 347.48 348.25 349.86 350.52 349.98
[341] 348.25 346.17 345.48 344.82 346.22 347.48 348.73 348.92 349.81 351.40
[351] 352.15 351.59 350.21 348.20 346.66 346.72 348.08 349.28 350.51 351.70
[361] 352.50 353.67 354.35 353.88 352.80 350.49 348.97 349.37 350.42 351.62
[371] 353.07 353.43 354.08 355.72 355.95 355.44 354.05 351.84 350.09 350.33
[381] 351.55 352.91 353.86 355.10 355.75 356.38 357.38 356.39 354.89 353.07
[391] 351.38 351.69 353.14 354.41 354.93 355.82 357.33 358.77 359.23 358.23

```

```

[401] 356.30 353.97 352.34 352.43 353.89 355.21 356.34 357.21 357.97 359.22
[411] 359.71 359.43 357.15 354.98 353.01 353.41 354.42 355.68 357.10 357.42
[421] 358.59 359.39 360.30 359.64 357.46 355.76 354.14 354.23 355.53 357.03
[431] 358.36 359.04 360.11 361.36 361.78 360.94 359.51 357.59 355.86 356.21
[441] 357.65 359.10 360.04 361.00 361.98 363.44 363.83 363.33 361.78 359.33
[451] 358.32 358.14 359.61 360.82 362.20 363.36 364.28 364.69 365.25 365.06
[461] 363.69 361.55 359.69 359.72 361.04 362.39 363.24 364.21 364.65 366.49
[471] 366.77 365.73 364.46 362.40 360.44 360.98 362.65 364.51 365.39 366.10
[481] 367.36 368.79 369.56 369.13 367.98 366.10 364.16 364.54 365.67 367.30
[491] 368.35 369.28 369.84 371.15 371.12 370.46 369.61 367.06 364.95 365.52
[501] 366.88 368.26 369.45 369.71 370.75 371.98 371.75 371.87 370.02 368.27
[511] 367.15 367.18 368.53 369.83 370.76 371.69 372.63 373.55 374.03 373.40
[521] 371.68 369.78 368.34 368.61 369.94 371.42 372.70 373.37 374.30 375.19
[531] 375.93 375.69 374.16 372.03 370.93 370.73 372.43 373.98 375.07 375.82
[541] 376.64 377.92 378.78 378.46 376.88 374.57 373.34 373.31 374.84 376.17
[551] 377.17 378.05 379.06 380.54 380.80 379.87 377.65 376.17 374.43 374.63
[561] 376.33 377.68 378.63 379.91 380.95 382.48 382.64 382.40 380.93 378.93
[571] 376.89 377.19 378.54 380.31 381.58 382.40 382.86 384.80 385.22 384.24
[581] 382.65 380.60 379.04 379.33 380.35 382.02 383.10 384.12 384.81 386.73
[591] 386.78 386.33 384.73 382.24 381.20 381.37 382.70 384.19 385.78 386.06
[601] 386.28 387.33 388.78 387.99 386.61 384.32 383.41 383.21 384.41 385.79
[611] 387.17 387.70 389.04 389.76 390.36 389.70 388.25 386.29 384.95 384.64
[621] 386.23 387.63 388.91 390.41 391.37 392.67 393.21 392.38 390.41 388.54
[631] 387.03 387.43 388.87 389.99 391.50 392.05 392.80 393.44 394.41 393.95
[641] 392.72 390.33 389.28 389.19 390.48 392.06 393.31 394.04 394.59 396.38
[651] 396.93 395.91 394.56 392.59 391.32 391.27 393.20 394.57 395.78 397.03
[661] 397.66 398.64 400.02 398.81 397.51 395.39 393.72 393.90 395.36 397.03
[671] 398.04 398.27 399.91 401.51 401.96 401.43 399.38 397.32 395.64 396.29
[681] 397.55 399.15 400.18 400.55 401.74 403.35 404.15 402.97 401.46 399.11
[691] 397.82 398.49 400.27 402.06 402.73 404.25 405.06 407.60 407.90 406.99
[701] 404.59 402.45 401.23 401.79 403.72 404.64 406.36 406.66 407.53 409.22
[711] 409.89 409.08 407.33 405.32 403.57 403.82 405.31 407.00 408.15 408.52
[721] 409.59 410.45 411.44 410.99 408.90 407.16 405.71 406.19 408.21 409.27
[731] 411.03 411.96 412.18 413.54 414.86 414.16 411.97 410.18 408.76 408.75
[741] 410.48 411.98 413.61 414.34 414.74 416.45 417.31 416.60 414.62 412.78
[751] 411.52 411.51 413.12 414.26 415.52 416.75 417.64 419.05 419.13 418.94
[761] 416.96

```

```
summary(avg_co2)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
312.4	329.4	353.4	356.4	380.5	419.1

Pre-industrial concentration was 280 ppm, so the percentage is `average / 280 * 100`.

```
co2_pct <- avg_co2 / 280 * 100
summary(co2_pct)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
111.6	117.6	126.2	127.3	135.9	149.7

The `if/else` block follows the `percentage` flag. Both settings are printed below so the answer key shows each branch. In the assignment itself, change `percentage` and run the same block again.

```
percentage <- TRUE

if (percentage == TRUE) {
  summary(co2_pct)
} else {
  summary(avg_co2)
}
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
111.6	117.6	126.2	127.3	135.9	149.7

```
percentage <- FALSE

if (percentage == TRUE) {
  summary(co2_pct)
} else {
  summary(avg_co2)
}
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
312.4	329.4	353.4	356.4	380.5	419.1

`date_decimal()` converts a decimal date to a calendar date. The three-letter time zone for Hawaii is "HST".

```
co2$date
```

[1] 1958.203 1958.288 1958.370 1958.455 1958.537 1958.622 1958.707 1958.789
 [9] 1958.874 1958.956 1959.041 1959.126 1959.203 1959.288 1959.370 1959.455
 [17] 1959.537 1959.622 1959.707 1959.789 1959.874 1959.956 1960.041 1960.126
 [25] 1960.205 1960.290 1960.372 1960.456 1960.538 1960.623 1960.708 1960.790
 [33] 1960.874 1960.956 1961.041 1961.126 1961.203 1961.288 1961.370 1961.455
 [41] 1961.537 1961.622 1961.707 1961.789 1961.874 1961.956 1962.041 1962.126
 [49] 1962.203 1962.288 1962.370 1962.455 1962.537 1962.622 1962.707 1962.789
 [57] 1962.874 1962.956 1963.041 1963.126 1963.203 1963.288 1963.370 1963.455
 [65] 1963.537 1963.622 1963.707 1963.789 1963.874 1963.956 1964.041 1964.126
 [73] 1964.205 1964.290 1964.372 1964.456 1964.538 1964.623 1964.708 1964.790
 [81] 1964.874 1964.956 1965.041 1965.126 1965.203 1965.288 1965.370 1965.455
 [89] 1965.537 1965.622 1965.707 1965.789 1965.874 1965.956 1966.041 1966.126
 [97] 1966.203 1966.288 1966.370 1966.455 1966.537 1966.622 1966.707 1966.789
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 [113] 1967.537 1967.622 1967.707 1967.789 1967.874 1967.956 1968.041 1968.126
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 [161] 1971.537 1971.622 1971.707 1971.789 1971.874 1971.956 1972.041 1972.126
 [169] 1972.205 1972.290 1972.372 1972.456 1972.538 1972.623 1972.708 1972.790
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 [193] 1974.203 1974.288 1974.375 1974.458 1974.542 1974.625 1974.708 1974.792
 [201] 1974.875 1974.958 1975.042 1975.125 1975.208 1975.292 1975.375 1975.458
 [209] 1975.542 1975.625 1975.708 1975.792 1975.875 1975.958 1976.042 1976.125
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 [257] 1979.542 1979.625 1979.708 1979.792 1979.875 1979.958 1980.042 1980.125
 [265] 1980.208 1980.292 1980.375 1980.458 1980.542 1980.625 1980.708 1980.792
 [273] 1980.875 1980.958 1981.042 1981.125 1981.208 1981.292 1981.375 1981.458
 [281] 1981.542 1981.625 1981.708 1981.792 1981.875 1981.958 1982.042 1982.125
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 [305] 1983.542 1983.625 1983.708 1983.792 1983.875 1983.958 1984.042 1984.125
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[345] 1986.875 1986.958 1987.042 1987.125 1987.208 1987.292 1987.375 1987.458
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 [361] 1988.208 1988.292 1988.375 1988.458 1988.542 1988.625 1988.708 1988.792
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 [425] 1993.542 1993.625 1993.708 1993.792 1993.875 1993.958 1994.042 1994.125
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 [441] 1994.875 1994.958 1995.042 1995.125 1995.208 1995.292 1995.375 1995.458
 [449] 1995.542 1995.625 1995.708 1995.792 1995.875 1995.958 1996.042 1996.125
 [457] 1996.208 1996.292 1996.375 1996.458 1996.542 1996.625 1996.708 1996.792
 [465] 1996.875 1996.958 1997.042 1997.125 1997.208 1997.292 1997.375 1997.458
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 [481] 1998.208 1998.292 1998.375 1998.458 1998.542 1998.625 1998.708 1998.792
 [489] 1998.875 1998.958 1999.042 1999.125 1999.208 1999.292 1999.375 1999.458
 [497] 1999.542 1999.625 1999.708 1999.792 1999.875 1999.958 2000.042 2000.125
 [505] 2000.208 2000.292 2000.375 2000.458 2000.542 2000.625 2000.708 2000.792
 [513] 2000.875 2000.958 2001.042 2001.125 2001.208 2001.292 2001.375 2001.458
 [521] 2001.542 2001.625 2001.708 2001.792 2001.875 2001.958 2002.042 2002.125
 [529] 2002.208 2002.292 2002.375 2002.458 2002.542 2002.625 2002.708 2002.792
 [537] 2002.875 2002.958 2003.042 2003.125 2003.208 2003.292 2003.375 2003.458
 [545] 2003.542 2003.625 2003.708 2003.792 2003.875 2003.958 2004.042 2004.125
 [553] 2004.208 2004.292 2004.375 2004.458 2004.542 2004.625 2004.708 2004.792
 [561] 2004.875 2004.958 2005.042 2005.125 2005.208 2005.292 2005.375 2005.458
 [569] 2005.542 2005.625 2005.708 2005.792 2005.875 2005.958 2006.042 2006.125
 [577] 2006.208 2006.292 2006.375 2006.458 2006.542 2006.625 2006.708 2006.792
 [585] 2006.875 2006.958 2007.042 2007.125 2007.208 2007.292 2007.375 2007.458
 [593] 2007.542 2007.625 2007.708 2007.792 2007.875 2007.958 2008.042 2008.125
 [601] 2008.208 2008.292 2008.375 2008.458 2008.542 2008.625 2008.708 2008.792
 [609] 2008.875 2008.958 2009.042 2009.125 2009.208 2009.292 2009.375 2009.458
 [617] 2009.542 2009.625 2009.708 2009.792 2009.875 2009.958 2010.042 2010.125
 [625] 2010.208 2010.292 2010.375 2010.458 2010.542 2010.625 2010.708 2010.792
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 [657] 2012.875 2012.958 2013.042 2013.125 2013.208 2013.292 2013.375 2013.458
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 [681] 2014.875 2014.958 2015.042 2015.125 2015.208 2015.292 2015.375 2015.458

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[689] 2015.542 2015.625 2015.708 2015.792 2015.875 2015.958 2016.042 2016.125
[697] 2016.208 2016.292 2016.375 2016.458 2016.542 2016.625 2016.708 2016.792
[705] 2016.875 2016.958 2017.042 2017.125 2017.208 2017.292 2017.375 2017.458
[713] 2017.542 2017.625 2017.708 2017.792 2017.875 2017.958 2018.042 2018.125
[721] 2018.208 2018.292 2018.375 2018.458 2018.542 2018.625 2018.708 2018.792
[729] 2018.875 2018.958 2019.042 2019.125 2019.208 2019.292 2019.375 2019.458
[737] 2019.542 2019.625 2019.708 2019.792 2019.875 2019.958 2020.042 2020.125
[745] 2020.208 2020.292 2020.375 2020.458 2020.542 2020.625 2020.708 2020.792
[753] 2020.875 2020.958 2021.042 2021.125 2021.208 2021.292 2021.375 2021.458
[761] 2021.542
```

```
date_decimal(co2$date, tz = "HST")
```

```
[1] "1958-03-15 23:39:07 HST" "1958-04-16 00:15:07 HST"
[3] "1958-05-16 00:19:26 HST" "1958-06-16 00:02:52 HST"
[5] "1958-07-16 00:07:12 HST" "1958-08-15 23:50:38 HST"
[7] "1958-09-15 23:34:04 HST" "1958-10-15 23:38:23 HST"
[9] "1958-11-16 00:14:24 HST" "1958-12-16 00:18:43 HST"
[11] "1959-01-16 00:02:09 HST" "1959-02-15 23:45:35 HST"
[13] "1959-03-15 23:39:07 HST" "1959-04-16 00:15:07 HST"
[15] "1959-05-16 00:19:26 HST" "1959-06-16 00:02:52 HST"
[17] "1959-07-16 00:07:12 HST" "1959-08-15 23:50:38 HST"
[19] "1959-09-15 23:34:04 HST" "1959-10-15 23:38:23 HST"
[21] "1959-11-16 00:14:24 HST" "1959-12-16 00:18:43 HST"
[23] "1960-01-16 00:08:38 HST" "1960-02-16 00:08:55 HST"
[25] "1960-03-15 23:50:29 HST" "1960-04-15 23:50:47 HST"
[27] "1960-05-16 00:08:03 HST" "1960-06-16 00:08:21 HST"
[29] "1960-07-16 00:25:37 HST" "1960-08-16 00:25:55 HST"
[31] "1960-09-16 00:26:12 HST" "1960-10-15 23:50:47 HST"
[33] "1960-11-15 23:51:04 HST" "1960-12-16 00:08:21 HST"
[35] "1961-01-16 00:02:09 HST" "1961-02-15 23:45:35 HST"
[37] "1961-03-15 23:39:07 HST" "1961-04-16 00:15:07 HST"
[39] "1961-05-16 00:19:26 HST" "1961-06-16 00:02:52 HST"
[41] "1961-07-16 00:07:12 HST" "1961-08-15 23:50:38 HST"
[43] "1961-09-15 23:34:04 HST" "1961-10-15 23:38:23 HST"
[45] "1961-11-16 00:14:24 HST" "1961-12-16 00:18:43 HST"
[47] "1962-01-16 00:02:09 HST" "1962-02-15 23:45:35 HST"
[49] "1962-03-15 23:39:07 HST" "1962-04-16 00:15:07 HST"
[51] "1962-05-16 00:19:26 HST" "1962-06-16 00:02:52 HST"
[53] "1962-07-16 00:07:12 HST" "1962-08-15 23:50:38 HST"
[55] "1962-09-15 23:34:04 HST" "1962-10-15 23:38:23 HST"
[57] "1962-11-16 00:14:24 HST" "1962-12-16 00:18:43 HST"
```

[59] "1963-01-16 00:02:09 HST" "1963-02-15 23:45:35 HST"
 [61] "1963-03-15 23:39:07 HST" "1963-04-16 00:15:07 HST"
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[759] "2021-05-17 21:00:00 HST" "2021-06-17 06:42:28 HST"  
[761] "2021-07-17 17:17:31 HST"
```

Question 3

```
for (i in 1:10) {  
  print(i * 3)  
}
```

```
[1] 3  
[1] 6  
[1] 9  
[1] 12  
[1] 15  
[1] 18  
[1] 21  
[1] 24  
[1] 27  
[1] 30
```

Question 4

```
i <- 1
while (i <= 10) {
  print(i * 3)
  i <- i + 1
}
```

```
[1] 3
[1] 6
[1] 9
[1] 12
[1] 15
[1] 18
[1] 21
[1] 24
[1] 27
[1] 30
```

The counter has to increase inside the loop. Without `i <- i + 1`, `i` stays at 1 and the loop never ends.

Question 5

This answer is your own profile. It should look like `https://github.com/your-username`.

Question 6

```
git pull
```

You have been away from the project, so the first step is to download the commits your teammates pushed while you were gone. Pulling first keeps you from editing an outdated copy of the files.

Question 7

```
git push
```

A commit only saves the change on your computer. Your teammate cannot see it until that commit is pushed to the shared remote repository.